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StrataFlux: Memory-Hierarchy-Native Low-Bit LLM Inference

Using Host DRAM and NVMe SSD as Precision Strata Recovers Near-Full Accuracy at 2-Bit Weight Quantization Without Extra GPU Memory

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StrataFlux recovers accuracy lost to 2–4-bit quantization by treating the full memory hierarchy—GPU HBM, host DRAM, and NVMe SSD—as a stratified precision resource. On 2-bit Llama-3.1-8B-Instruct, WikiText-2 perplexity drops from 9.46 to 8.40 at 95% base throughput; on 2-bit Qwen3-8B, AIME 2024/2025 accuracy rises from 11.0% to 47.1% with no additional GPU memory.

AT A GLANCE

Problem: 2-bit quantization reduces GPU memory but collapses accuracy; existing mixed-precision fixes keep all higher-precision weights on-GPU, negating the memory benefit.

Why it matters: Enabling 2-bit inference at acceptable accuracy would halve peak GPU memory for many deployments, making 70B+ models accessible on single-GPU servers.

Method: Stratified sensitivity profiling assigns weight layers to GPU HBM (2-bit), DRAM (4-bit), or NVMe SSD (8-bit); a prefetch scheduler hides tier-B/C transfer latency behind GPU compute.

Result: 2-bit Llama-3.1-8B perplexity 8.40 (vs. 9.46 baseline); 2-bit Qwen3-8B AIME accuracy 47.1% (vs. 11.0%).

Evidence: WikiText-2, AIME 2024/2025, 128K long-context; compared against 2-bit RTN, GPTQ, AWQ; RTX 5090.

The Big Picture

Large language model inference is bounded by memory bandwidth, not compute. Quantization reduces both peak memory and bandwidth requirements: 2-bit weights need 16× less memory than FP32. The problem is accuracy. At 2 bits, quantization error is large enough to damage reasoning ability, particularly on tasks requiring precise recall or long multi-step chains.

Prior solutions (GPTQ, AWQ, SpQR) identify sensitive weights and store them at higher precision *on the GPU*—but this cancels the memory benefit for the sensitive fraction. StrataFlux reframes the question: host DRAM is 8–32× larger and 5–10× slower; NVMe SSD is larger still. Sensitive weights that need 4- or 8-bit precision can live in DRAM or SSD and be streamed to GPU only when needed, overlapped with GPU computation on the previous layer.

Why Existing Approaches Fall Short

Uniform quantization (RTN, 2-bit) is fast and memory-efficient but loses too much accuracy in sensitive layers, especially attention projections and early embedding layers.

Mixed-precision on-GPU (GPTQ, AWQ, SpQR) keeps sensitive weights at 4–8 bits on the GPU. This recovers accuracy but requires extra GPU memory, partially defeating the purpose of quantization.

CPU offloading (FlexGen, DejaVu) moves entire layers to CPU, introducing large latency spikes. StrataFlux differs: it offloads only the *higher-precision copies* of sensitive layers, not the full layer; the base 2-bit copy always stays on GPU for fast path.

Core Mechanism: Three-Tier Precision

Stratum A (GPU HBM): Base quantized weights of all layers at the target bit-width. Always resident.

Stratum B (host DRAM): 4-bit copies of the top-k% most sensitive layers, paged in one block ahead of the layer it corrects.

Stratum C (NVMe SSD): 8-bit copies of an extremely sensitive subset (typically first and last 2 transformer blocks), paged asynchronously via direct DMA.

For a stratum-B/C layer, the forward pass combines the base 2-bit GPU weight with a residual correction loaded from DRAM or SSD: `y = W_base x + ΔW_tier x` where $\Delta W_{tier} = W_{tier} - W_{base}$. This residual form ensures an approximate result if the DMA transfer has not completed, instead of blocking.

Technical Formulation

Sensitivity of layer l is estimated by the Fisher trace:

```
S_l ≈ (1/|C|) ∑_{x∈C} ||W_l x − Ŵ_l x||^2
```

where C is a calibration corpus, W_l is full-precision weight, and $\hat{W}_l$ is 2-bit quantized. Layers are ranked by S_l and assigned by thresholds θ_B and θ_C, set by hardware memory budget constraints.

Learning or Inference Procedure

StrataFlux requires no fine-tuning. The workflow: (1) Run calibration pass on 128 sequences to compute S_l; (2) Assign layers to strata; (3) Quantize and place stratum-B copies in DRAM, stratum-C in SSD; (4) At inference, the prefetch scheduler issues DMA requests for the next 3 stratum-B layers and 1 stratum-C layer ahead of execution, overlapping transfers with GPU computation.

What the Guarantee Says

No theoretical accuracy bound is provided. The empirical claim is Pareto-dominance over existing mixed-precision methods: StrataFlux achieves lower perplexity at lower GPU memory than any configuration of GPTQ or AWQ tested. No latency bound under SSD-latency spikes is given; the base path ensures correctness but not accuracy.

Experimental Findings

- **Perplexity:** 2-bit Llama-3.1-8B WikiText-2 PPL 8.40 (StrataFlux) vs. 9.46 (2-bit RTN base); 95% base throughput on RTX 5090.
- **Reasoning:** 2-bit Qwen3-8B AIME 2024/2025 accuracy 47.1% vs. 11.0% (2-bit RTN); GPU memory unchanged.
- **Long-context:** 3-bit weights + 2-bit KV at 128K context matches vLLM 4-bit AWQ + 8-bit KV at half the GPU memory.
- **Throughput:** Prefetch scheduler achieves 95% of base 2-bit throughput by hiding DRAM latency.

Ablations and Interpretation

Stratum-C necessity: Removing SSD tier and assigning stratum-C layers to DRAM at 4-bit increases DRAM footprint by 18% at no accuracy change.

Residual form: Replacing residual correction with full replacement reduces throughput by 12% because the base 2-bit GPU path is always bypassed.

Calibration size: 32 sequences achieve 94% of sensitivity ordering quality of 512 sequences; 128 reaches 99%.

Model size: Relative perplexity improvement is larger for 8B models (1.1 PPL points) than 70B models (0.4 PPL points).

Reference

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